

Total No. of Questions : 10]

SEAT No. :

P34

[Total No. of Pages : 3

[5871]-534

B.E. (Mechanical)

DYNAMICS OF MACHINERY
(2015 Pattern) (Semester- I) (402043)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data if necessary.
- 5) Figures to the right indicate full marks.

- Q1)** a) Explain Dry friction damping. [4]
b) Explain the Principal Modes of vibration. [6]

OR

- Q2)** a) Explain frequency response curve with neat labeled diagram. [5]
b) A vibratory system in a vehicle is to be designed with the following parameter.
 $K = 100\text{N/m}$, $C = 2\text{ N-sec/m}$, $m=1\text{ kg}$.
Calculate the decrease of amplitude from its starting value after 3 complete oscillation and frequency of oscillation. [5]

- Q3)** a) Explain with neat diagram mathematical model of a motorbike. [4]
b) Explain following term : [6]
i) Vibration isolation
ii) Quality factor
ii) Magnification factor

OR

- Q4)** a) Explain following terms : [4]
i) Zero frequency deflection
ii) Node point

P.T.O.

- b) A vehicle moves over a road surface having approximately the sinusoidal profile with a wavelength of 10 m and amplitude of 80 mm. The vehicle is moving with a velocity of 55 km/hr. Calculate critical speed of vehicle, if the amplitude of vibration is 25 mm and mass of vehicle is 500 kg. [6]

OR

- Q5) a)** A five cylinder in-line engine running at 750 rpm. has successive cranks 144° apart, the distance between the cylinder center lines being 375mm. The piston stroke is 225 mm and the ratio of the connecting rod to the crank is 4. Examine the engine for balance of primary and secondary forces and couples. Find the maximum values of these and the position of the central crank at which these maximum values occur. The reciprocating mass for each cylinder is 15 kg. [14]
- b) Explain direct & reverse crank method with neat diagram. [4]

OR

- Q6) a)** The axes of three cylinder air compressor are at 120° to one another and their connecting rods are coupled to single crank. The length of connecting rod is 240 mm and the stroke is 160 mm. The reciprocating parts have a mass of 24kg per cylinder. Determine the primary and secondary forces if the engine runs at 2000 rpm. Use the concept of direct and reverse crank. [12]
- b) Explain the partial balancing of single cylinder engine [6]
- Q7) a)** Explain various parameters used to measure vibratory response of a vibrating system. [6]
- b) The static deflection of vibrometer mass is 20 mm. The instrument when attached to a machine vibrating with a frequency of 125 cpm records relative amplitude of 0.03 mm find [6]
1. Amplitude of vibration
 2. Maximum velocity of vibration.
 3. Maximum acceleration
- c) Explain vibration isolators. [4]

OR

- Q8) a)** Explain MR damper as an effective control over vibration. [6]
b) It is required to measure the maximum acceleration of a machine, which vibrates violently with the frequency of 700 cycles per min. Accelerometer with negligible damping is attached to it and the indicator travels by 8.2mm. If the accelerometer weighs 0.5 kg and has a spring rate of 17500 N/m, what is the maximum amplitude and maximum acceleration of the part? [4]
c) Explain FFT analyzer with neat schematic diagram. [6]

OR

- Q9) a)** Noise at construction site is contributed by a few construction activities such as piling work: 104dB, bulldozer: 94 dB, scraper: 93 dB, Mobile compressor: 73 dB and mechanical shovel: 76 dB. What is overall sound pressure level? [6]
 Also determine sound pressure level when bulldozer is not is working condition.
b) What is sound field? Explain various types of sound fields in the vicinity of sound source. [6]
c) Explain sound level meter with neat diagram. [4]

OR

- Q10)a)** Show that if distance from point source is doubles, the sound intensity level decreases by 6 dB. [6]
b) Explain in brief various sources of noise and how to control the same. [6]
c) Determine the sound pressure level of a source that generates a following rms sound pressure. [4]
 a) 1.5 N/m^2
 b) 0.6 Pa

